

Output tables for the test of Multiple comparisons.

March 1, 2025

1 Average rankings of Friedman test

Average ranks obtained by applying the Friedman procedure

Algorithm	Ranking
GWO	1.0934
SA	2.9444
HC	1.9621

Table 1: Average Rankings of the algorithms

Friedman statistic considering reduction performance (distributed according to chi-square with 2 degrees of freedom: 679.247475.

P-value computed by Friedman Test: 2.377553709465019E-10.

2 Post hoc comparisons

Results achieved on post hoc comparisons for $\alpha = 0.05$, $\alpha = 0.10$ and adjusted p-values.

2.1 P-values for $\alpha = 0.05$

i	algorithms	$z = (R_0 - R_i)/SE$	p
3	GWO vs. SA	26.046021	0
2	SA vs. HC	13.822513	0
1	GWO vs. HC	12.223508	0

Table 2: P-values Table for $\alpha = 0.05$

2.2 P-values for $\alpha = 0.10$

i	algorithms	$z = (R_0 - R_i)/SE$	p
3	GWO vs. SA	26.046021	0
2	SA vs. HC	13.822513	0
1	GWO vs. HC	12.223508	0

Table 3: P-values Table for $\alpha = 0.10$

2.3 Adjusted p -values

i	hypothesis	unadjusted p
1	GWO vs .SA	0
2	SA vs .HC	0
3	GWO vs .HC	0

Table 4: Adjusted p -values